

Week 15: Personal Project Part 2

Overview

This assignment turns Week 15 concepts into a small, finishable Java program or project milestone. Start from the provided template, preserve the public class filename, and run the program in IntelliJ before submitting.

Objectives

- Integrate validation, exceptions, regular expressions, and file persistence.
- Deliver a dependable program with clear run instructions.
- Explain and demonstrate the completed design.

Project work week

No new Think Java chapter is assigned for this project milestone.

Optional supplemental reading - previously covered chapters

- [Chapter 5 - Conditionals and Logic](#)
- [Chapter 6 - Loops and Strings](#)
- [Chapter 11 - Designing Classes](#)

Complete these steps

- Copy your working Part 1 project into a new IntelliJ project or a clearly labeled Part 2 folder before making changes.
- Keep the original purpose working while adding the requirements below.
- Use meaningful String manipulation that changes, searches, formats, or analyzes project text.
- Use a loop to keep asking until at least one important user input is valid.
- Use try/catch to handle a realistic failure and explicitly throw at least one exception with the throw keyword.
- Use Pattern and Matcher, or String.matches, with at least one regular expression that serves the project's purpose.
- Load project data from a file and save project data to a file; use a portable relative path.
- Display helpful messages when input or file operations fail instead of ending with an unexplained stack trace.
- Preserve the foundational Part 1 requirements unless your approved design makes one unnecessary and your instructor approves the change.
- Test the complete project from a clean start, including invalid input and a missing or empty data file.
- Update README.md with every required file, exact OpenJDK 26 and IntelliJ run steps, the data-file location, and sample input/output.
- Record a 5-10 minute presentation explaining the design changes, demonstrating the full workflow, and identifying the Java concepts used.
- Submit all Java files, required data files, README.md, and the presentation in the locations specified by your instructor.

What will be checked

Use this list as your final review before submitting.

- 20 points: the expanded project runs and completes its main purpose without major errors.
- 15 points: presentation content clearly explains design, functionality, and Java concepts.
- 15 points: presentation delivery is organized, concise, and understandable.
- 10 points: readable organization and naming.
- 10 points: meaningful comments.
- 10 points: file input and output correctly load and save project data.
- 5 points each: meaningful String manipulation, loop-based input validation, try/catch plus an explicitly thrown exception, and a useful regular expression.
- The submitted file contains no unfinished TODO marker from the starter template.